**McGill University**

MGSC 416 | Data Driven Models for Operation Analysis

Group Project Report

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# Optimizing Mercedes F1 Race Strategy at the 2025 Canadian Grand Prix

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Professor Rim Harris

## I. Introduction

Formula 1 is a discipline where each team operates under very different budgets, with different levels of car performance and where drivers bring different driving styles and abilities. Yet despite those structural differences, race strategy is one of the few dimensions that can level the playing field, and a good strategy can sometimes win races where pure pace cannot. In modern F1, race strategy mostly revolves around tire management and pit stop timing, and where the core decisions are straightforward: when to stop, and which compound tire to fit. However the subtlety of those models lies in the vast amount of parameters to account for, which will determine the correct sequencing of those pit stop: for example whether to pit early and run fresh tyres in response to changing weather, or to wait and react to a rival's stop to gain positions without ever needing to pass on the track. A slow wheel change or a mechanical error can cost several seconds and turn a theoretically winning strategy into a losing one. This is another factor to consider because pit stop execution varies greatly across teams. To solve this issue dynamically over the course of a race weekend, elite Formula One teams actually rely on advanced real-time simulation software and enormous computational resources.This project attempts to capture the core logic and formulation of that process using course concepts.

More specifically, this project focuses on optimizing the race strategy for the two Mercedes drivers, Andrea Kimi Antonelli (ANT) and George Russell (RUS), during the 2025 Canadian Grand Prix in Montreal. The Canadian Grand Prix is particularly interesting because the Circuit Gilles-Villeneuve is a stop-and-go track with high tire degradation, many safety-car periods, and limited overtaking opportunities, making pit-stop timing especially consequential. On such a circuit, the difference between a well-timed stop and a poorly-timed one can easily translate into several positions gained or lost, therefore making it an ideal setting to explore the value of a structured, data-driven approach to strategy.

The optimization problem we are formulating is, at its core, a mixed-integer program. We seek to minimize the total race time for each driver, by deciding on each lap which tire compound to use, when to pit, and how many stints to complete. The constraints reflect the real-world rules and physical realities of the sport: F1 regulations mandate that each driver must use at least two different tire compounds during a race, pit stops incur a fixed time penalty, and tires degrade over time in a way that increases lap times as they age. These elements together create a non-trivial combinatorial structure, where the interaction between compound choice, stint length, and pit stop timing must all be jointly optimized.

The analytical approach combines two components: first, a predictive step where we use observed lap time data from FastF1 to estimate how lap times evolve as a function of tire age and compound, and second, a prescriptive step where we embed those estimated parameters into a Gurobi-based integer program to find the optimal pit stop strategy. This dual structure, like prediction feeding into optimization, is precisely the kind of integrated data-driven modeling that the course has emphasized, and it allows us to ground our prescriptive decisions in real empirical evidence rather than arbitrary assumptions.

## I. Data Overview

The data used in this project comes from FastF1, an open-source Python package developed by Oehrly and available on GitHub. FastF1 provides programmatic access to official Formula 1 timing and

telemetry data by querying the F1 live timing API as well as the Ergast-compatible jolpica-f1 API, which together cover race results, session data, lap-by-lap timing, tire information, and car telemetry for all sessions from 2018 onward. All data returned by FastF1 is structured as Pandas DataFrames, which makes it easy to manipulate and feed directly into analytical pipelines. The package also implements a local caching system so that repeated queries do not re-download the same data, which is convenient for iterative development.

For this project, we load the race session of the 2025 Canadian Grand Prix using the command fastf1.get_session(2025, 'Montreal', 'R'), and we extract lap-level data for the two Mercedes drivers, Antonelli and Russell. For each driver and each lap, we retrieve four variables: the lap number, the lap time in seconds, the tire compound used (Soft, Medium, or Hard), and the tire life expressed as the number of laps already completed on that set of tires. These four variables constitute the core dataset on which both the predictive and prescriptive components of the analysis are built.

The raw data requires some minimal cleaning. Lap times are initially stored as Timedelta objects and must be converted to floating-point seconds. Laps with missing lap time values, which typically correspond to in-laps, out-laps, or laps affected by yellow flags or safety car periods, are dropped from the dataset before regression to avoid distorting the tire degradation estimates. After cleaning, the dataset contains roughly 60 to 70 usable laps per driver, which is sufficient to estimate compound-specific degradation curves. One notable data wrangling step is the need to filter out the first lap of each race, since lap one times are heavily influenced by the standing start and grid position rather than by tire performance; hence that lap is excluded from the regression training set as well.

A few summary statistics are worth noting to motivate the methodology. Across both drivers, median lap times on Medium tires hover around 76 to 77 seconds at the beginning of a stint, and they increase by roughly 0.05 to 0.10 seconds per additional lap of tire age, reflecting a moderate but consistent degradation rate. Soft tires appear faster in the early laps of a stint but degrade more steeply, while Hard tires show a flatter degradation profile but a higher base time. These patterns directly inform how the expected lap time function is constructed in the optimization model, and they confirm that the data, even in this relatively compact form, contains enough signal to meaningfully differentiate tire strategies.

## II. Methodology

Our approach combines predictive modeling and optimization to determine the optimal tire and pit stop strategy for each driver. Let $d$ denote the driver (ANT or RUS), $i = 1, \ldots, 70$ the race laps, $c \in \{\text{SOFT, MEDIUM, HARD}\}$ the tire compound and $a$, tire age. Race data from the Canadian Grand Prix is used to estimate lap-time behavior. Data from FP1, FP2, FP3 and qualifying sessions provides the basis for approximating each driver's available tire inventory. To ensure data quality, we remove laps faster than 107% of the driver's fastest lap. Tire availability is reconstructed by starting from the standard dry allocation and tracking both unused sets and previously used sets with their corresponding tire age based on practice and qualifying sessions.

We begin with a pooled regression model where lap time depends on tire age, lap number,

compound, and interaction terms. One of the main problems we encountered when trying to estimate tire degradation rate is that tire age and lap number increase at exactly the same rate, making them perfectly collinear. Hence, our first version could not separate the effect of tire wear from the effect of fuel burn (because the car gets faster throughout the race as fuel burns off, this biases our degradation signal) our earlier models even produced negative degradation estimates for durable compounds like Hard. To address this multicollinearity, the final model separates the estimation of degradation and base pace. Tire degradation rates are estimated using a panel regression across multiple 8 races from the 2025 season prior to the Montreal Grand Prix with tire age, compound interactions, lap number and race fixed effects. In this panel regression, lap number captures fuel burn and track evolution, while race fixed effects control for circuit-specific differences, allowing tire age to isolate true degradation. This approach assumes that degradation patterns are sufficiently transferable across circuits once race-specific effects are controlled for which is a necessary simplification given limited data availability. Base pace for each driver and compound is then calibrated using Montreal race data. In other words, we combined slopes from the panel of 8 races, and baseline values from Montreal's data only, making our predicted lap time function $T_{(d,\,i,\,c,\,a)}$ conceptually valid.

For each driver d and compound c, we estimate:

*LapTime(d,c,t) = alpha(d,c) + gamma(d,c) * TireLife(d,c,t)*

These coefficients are then used to construct predicted lap times:
*T(d,i,c,a) = alpha(d,c) + gamma(d,c) * a*

Here, gamma(d,c) represents the estimated tire degradation rate from the panel regression and alpha(d,c) represents the base pace calibrated using Montreal race data. These predicted values serve as inputs to the optimization model. Note that although the function is indexed by lap i, in the implementation it depends on driver, compound and tire age. When panel estimates are unavailable or negative, degradation rates are replaced with compound-level median values to ensure physically meaningful and non-negative tire wear. For combinations with insufficient data, particularly for soft tires, fallback values are used to avoid unrealistic predictions. It is important to note that driver-specific degradation estimates may also reflect factors beyond tire behavior. For example, ANT is in his first session and higher degradation may partially reflect experience effects rather than purely tire performance.

We formulate a mixed-integer optimization model to minimize total race time.

***The main decision variable is:***

*x(d,i,c,a) = 1* if driver d uses compound c with age a on lap i, 0 otherwise

***Additional variables include:***

*y(d,i) = 1* if driver d pits after lap i, 0 otherwise

$u(d,c) = 1$ if driver d uses compound c at least once, 0 otherwise
$z(d,i,c,a) = 1$ if a new tire set (c,a) is mounted on lap i

The objective is to minimize total race time of both drivers that reflects a team-level optimization rather than independent strategies. It is defined as the sum of predicted lap times and pit-stop penalties:

Minimize sum over $d,i,c,a\ [\ T(d,i,c,a) * x(d,i,c,a)\ ] + P * \text{sum over } d,i\ [\ y(d,i)\ ],$ where $P = 20$ seconds represents the time loss associated with a pit stop.

The model enforces realistic race dynamics through operational constraints. Each driver uses exactly one tire compound per lap and tire age increases by one unless a pit stop occurs in which case a new tire is mounted and age resets. To reflect F1 regulations, each driver must use at least two compounds. Tire usage is also limited by estimated availability from practice and qualifying to ensure feasible allocation. A team-level constraint prevents both drivers from pitting on the same lap. Finally, maximum tire-life limits are imposed for each compound to avoid unrealistic no-stop strategies.

## III. Results and Insights

The predictive model shows strong performance in explaining lap-time variation. The pooled regression achieves an in-sample $R^2$ of 0.8863 and an out-of-sample $R^2$ of 0.8619 with a low RMSE of 0.281 seconds and MAPE of 0.26%. These results indicate high predictive accuracy and good generalization to new data. The model also shows that lap number has a significant negative effect on lap time with a coefficient of −0.0329 and $p < 0.001$ which aligns with improvements due to fuel burn and track evolution. Tire life alone is not statistically significant, but its interaction with compounds is highly significant. This confirms that degradation depends strongly on tire type rather than increasing uniformly.

Panel regression results further reinforce this finding by showing clear differences in degradation across compounds. Soft tires exhibit the steepest degradation, medium tires show moderate degradation and hard tires are the most stable. This pattern is consistent across both drivers and supports the use of compound-specific degradation rates in the optimization model.

Using these predictions, the optimization model identifies a one-stop strategy as optimal for both drivers by achieving a total combined race time of 10,656.34 seconds with a negligible optimality gap (0.0059%). Both ANT and RUS follow nearly identical strategies, starting on hard tires for a long initial run spanning approximately laps 1 to 49 (RUS) or 50 (ANT) before switching to medium tires for the final portion of the race. The pit stops are staggered by one lap to satisfy team constraints.

The results highlight that hard tires dominate race strategy because their low degradation allows drivers to minimize time loss over long runs. Soft tires are not used at all which indicates that their higher degradation outweighs any short-term speed advantage. Medium tires are only used in the final segment because their moderate degradation is less costly over a shorter duration. The optimal pit timing around lap 50 reflects a balance between increasing degradation on aging hard tires and the fixed cost of a pit stop.

Overall, the findings suggest that under stable race conditions, tire durability is more valuable than short-term performance and a hard-to-medium one-stop strategy minimizes total race time, which seems pretty coherent with the reality of this race, because the low degradation of Hard tires at this circuit makes the cost of staying on the track far lower than the cost of an additional pit stop. In fact, the solution proposed by our model closely mirrors the actual strategy adopted by ANT and RUS during this exact same race: starting with Hard tires before switching to Medium then Soft at the very end of the race. The results also show that pit-stop costs play a bigger role in strategy than the relatively small lap-time differences across tire compounds.

## IV. Conclusion and Limitations

Our project applied a two-stage data-driven framework to optimize race strategy for the two Mercedes drivers Antonelli and Russel at the 2025 Canadian Grand Prix. By combining a panel drawn from data of 8 different races, we built a regression to estimate tire degradation, then a mixed-integer program to prescribe pit stop timing and compound selection. Our model outputs a one-stop Hard-to-Medium strategy for both drivers as the optimal solution, which almost matches the strategy executed by both drivers during the actual race. Beyond providing an optimal strategy, the model also gives us an interesting insight: at Montreal, Hard tire degradation is low enough that the cost of staying on the track without pitting, with a single late stop at the very end of the race, is the rational choice.

The main limitation is that our model assumes perfect clear race conditions and does not account for the uncertainty factor that we usually find in F1 races, for example safety car periods can reduce the effective pit cost from 20 seconds to near zero. Since Montreal is historically one of the highest safety car frequency circuits on the calendar, incorporating a probability factor to pit cost would be a logical extension to our current model. A natural extension would be to incorporate stochastic optimization or scenario-based modeling to explicitly account for safety car probabilities and their impact on pit stop timing decisions.

Furthermore, while our model is conceptually valid, real F1 strategy teams operate with much richer data, which is unavailable to the public. In fact, F1 teams run dedicated tire test sessions and have access to proprietary tire compound specifications. They also continuously update their models with live telemetry throughout the race weekend. To solve for our lack of data, we derived from eight other races across different circuits with different surfaces and conditions which may not fully capture Montreal-specific tire behavior. This limitation becomes apparent when analysing Russell's estimates, where the panel returned near-identical degradation slopes across all three compounds, which may indicate that his prior race data did not use enough compound variety to differentiate each of them meaningfully. The Hard compound degradation of roughly 0.02 seconds per lap is unreasonably low, even for the Montreal track, our model hence assumes that keeping Hard tire over laps is nearly costless, which in turn pushes the optimal pit stop very late into the race (around the 50th lap). Ideally, better data would give us better-calibrated rates and would likely produce a more realistic strategic solution.

## Appendix

### A. Tables:

**Table 1 - Tire Inventory Output**

| Driver | Soft - New | Soft Used | Med - New | Med - Used | Hard - New | Hard - Used |
|---|---|---|---|---|---|---|
| ANT | 0 | x1 - 1 lap old<br>x3 - 3 lap old | 0 | x1 - 3 lap old | x1 | x1 - 1 lap old |
| RUS | 0 | x3 - 3 lap old | 0 | x2 - 3 lap old | x1 | x1 - 1 lap old |

### B. Figures:

**Figure 1 and 2 - OLS Result for Antonelli and Russell**

```
--- Panel OLS: ANT ---
                            OLS Regression Results
==============================================================================
Dep. Variable:             LapTime_sec   R-squared:                       0.992
Model:                             OLS   Adj. R-squared:                  0.991
Method:                  Least Squares   F-statistic:                     3059.
Date:                 Wed, 01 Apr 2026   Prob (F-statistic):          3.63e-290
Time:                         22:15:06   Log-Likelihood:                -276.58
No. Observations:                  299   AIC:                             577.2
Df Residuals:                      287   BIC:                             621.6
Df Model:                           11
Covariance Type:             nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                     91.5145      0.202    452.558      0.000      91.117      91.913
TyreLife                   0.0239      0.004      5.743      0.000       0.016       0.032
TyreLife_x_MEDIUM          0.0280      0.007      3.944      0.000       0.014       0.042
TyreLife_x_SOFT            0.0643      0.011      5.720      0.000       0.042       0.086
LapNumber                 -0.0542      0.003    -17.615      0.000      -0.060      -0.048
Race_2025_Bahrain          7.4460      0.235     31.654      0.000       6.983       7.909
Race_2025_China            7.4033      0.194     38.136      0.000       7.021       7.785
Race_2025_Emilia_Romagna  -8.7674      0.204    -43.034      0.000      -9.168      -8.366
Race_2025_Japan            2.0808      0.206     10.097      0.000       1.675       2.486
Race_2025_Miami            1.7629      0.215      8.215      0.000       1.341       2.185
Race_2025_Monaco         -12.5318      0.231    -54.213      0.000     -12.987     -12.077
Race_2025_Saudi_Arabia     3.4249      0.200     17.152      0.000       3.032       3.818
==============================================================================
Omnibus:                       24.655   Durbin-Watson:                   0.808
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               49.165
Skew:                           0.444   Prob(JB):                     2.11e-11
Kurtosis:                       4.777   Cond. No.                         571.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
  HARD: raw panel slope = 0.02389 s/lap
  MEDIUM: raw panel slope = 0.05186 s/lap
  SOFT: raw panel slope = 0.08817 s/lap
```

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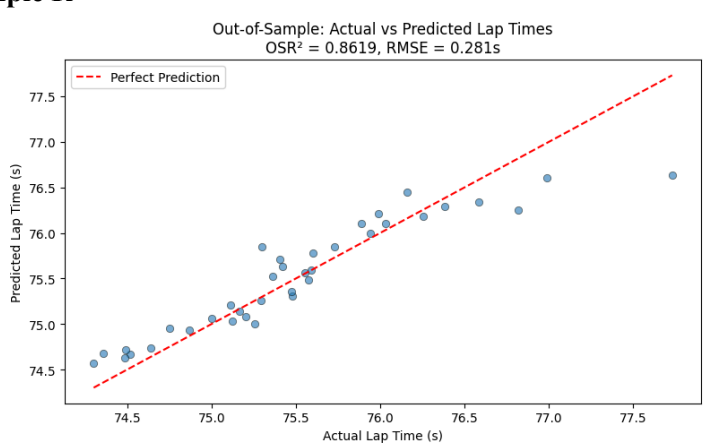
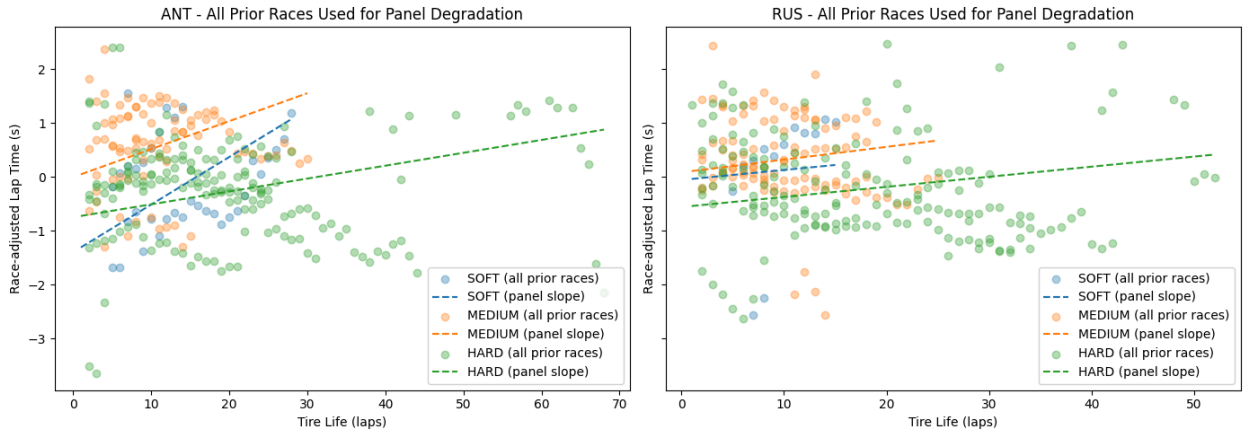
  HARD: raw panel slope = 0.02389 s/lap
  MEDIUM: raw panel slope = 0.05186 s/lap
  SOFT: raw panel slope = 0.08817 s/lap
```

**Figure 3 - Out-of Sample $R^2$**



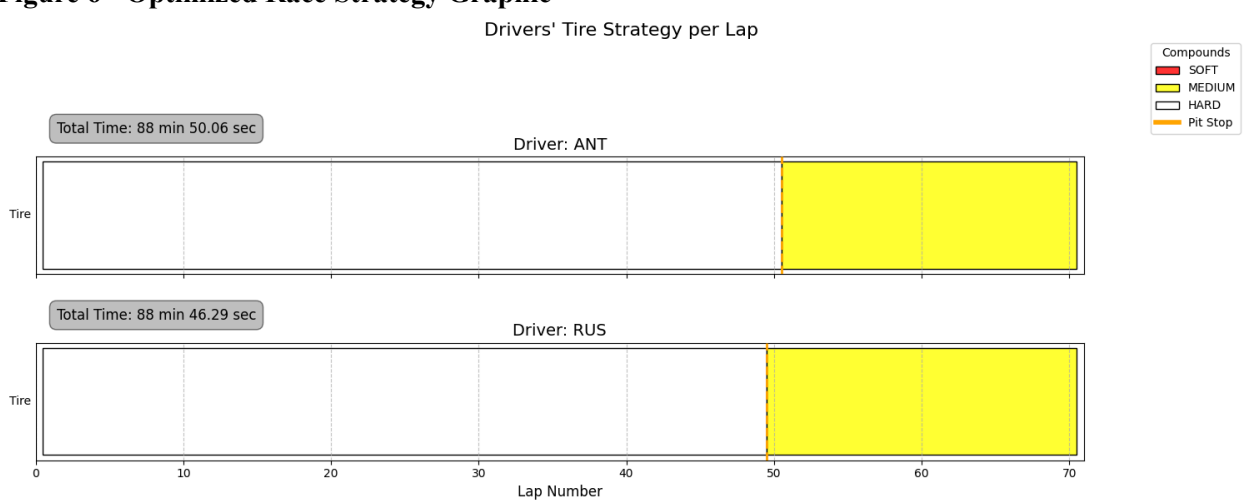
## Figure 4 - Adjusted Degradation curves




## Figure 5 - Race Strategy Output for Antonelli

```
--- Strategy for ANT ---
Lap 1: HARD (Age 1)
Lap 2: HARD (Age 2)
Lap 3: HARD (Age 3)
Lap 4: HARD (Age 4)
Lap 5: HARD (Age 5)
Lap 6: HARD (Age 6)
Lap 7: HARD (Age 7)
Lap 8: HARD (Age 8)
Lap 9: HARD (Age 9)
Lap 10: HARD (Age 10)
Lap 11: HARD (Age 11)
Lap 12: HARD (Age 12)
Lap 13: HARD (Age 13)
Lap 14: HARD (Age 14)
Lap 15: HARD (Age 15)
Lap 16: HARD (Age 16)
Lap 17: HARD (Age 17)
Lap 18: HARD (Age 18)
Lap 19: HARD (Age 19)
Lap 20: HARD (Age 20)
Lap 21: HARD (Age 21)
Lap 22: HARD (Age 22)
Lap 23: HARD (Age 23)
Lap 24: HARD (Age 24)
Lap 25: HARD (Age 25)
Lap 26: HARD (Age 26)
Lap 27: HARD (Age 27)
Lap 28: HARD (Age 28)
Lap 29: HARD (Age 29)
Lap 30: HARD (Age 30)
Lap 31: HARD (Age 31)
Lap 32: HARD (Age 32)
Lap 33: HARD (Age 33)
Lap 34: HARD (Age 34)
Lap 35: HARD (Age 35)
Lap 36: HARD (Age 36)
Lap 37: HARD (Age 37)
Lap 38: HARD (Age 38)
Lap 39: HARD (Age 39)
Lap 40: HARD (Age 40)
Lap 41: HARD (Age 41)
Lap 42: HARD (Age 42)
Lap 43: HARD (Age 43)
Lap 44: HARD (Age 44)
Lap 45: HARD (Age 45)
Lap 46: HARD (Age 46)
Lap 47: HARD (Age 47)
Lap 48: HARD (Age 48)
Lap 49: HARD (Age 49)
Lap 50: HARD (Age 50)
Lap 51: MEDIUM (Age 3)  <-- pitted after lap 50
Lap 52: MEDIUM (Age 4)
Lap 53: MEDIUM (Age 5)
Lap 54: MEDIUM (Age 6)
Lap 55: MEDIUM (Age 7)
Lap 56: MEDIUM (Age 8)
Lap 57: MEDIUM (Age 9)
Lap 58: MEDIUM (Age 10)
Lap 59: MEDIUM (Age 11)
Lap 60: MEDIUM (Age 12)
Lap 61: MEDIUM (Age 13)
Lap 62: MEDIUM (Age 14)
Lap 63: MEDIUM (Age 15)
Lap 64: MEDIUM (Age 16)
Lap 65: MEDIUM (Age 17)
Lap 66: MEDIUM (Age 18)
Lap 67: MEDIUM (Age 19)
Lap 68: MEDIUM (Age 20)
Lap 69: MEDIUM (Age 21)
Lap 70: MEDIUM (Age 22)
```

## Figure 6 - Optimized Race Strategy Graphic



## C. Mathematical Formulations

### 1. Tire Inventory

Standard Pirelli allocation:

$$N_0(SOFT) = 8, \quad N_0(MEDIUM) = 3, \quad N_0(HARD) = 2$$

Let U(d, c) = number of sets of compound c used by driver d in FP1, FP2, FP3, Q. Available sets for the race:

$$N_race(d, c) = N_0(c) - U(d, c)$$

Used sets carry initial age $a_0$ equal to their recorded max tire life. Final inventory:

$$Inv(d, c, a) = \text{number of sets of compound } c \text{ with initial age } a \text{ available to driver } d$$

### 2. Predictive Model

- **Pooled OLS Regression**

A pooled OLS is fitted on Montreal race data for both drivers:

$$LapTime = \beta_0 + \beta_1 \cdot TyreLife + \beta_2 \cdot LapNumber + \beta_3 \cdot \mathbb{1}_\{MED\} + \beta_4 \cdot \mathbb{1}_\{SOFT\} + \beta_5 \cdot (TyreLife \times \mathbb{1}_\{MED\}) + \beta_6 \cdot (TyreLife \times \mathbb{1}_\{SOFT\}) + \varepsilon$$

where 𝟙_{MED} and 𝟙_{SOFT} are indicator variables with HARD as baseline, and LapNumber captures fuel burn and track evolution.

- **Validation Metrics (70/30 split)**

$$OSR^2 = 1 - \Sigma_i(\hat{y}_i - y_i)^2 / \Sigma_i(y_i - \bar{y})^2$$

$$RMSE = \sqrt{[\ (1/n) \sum_i(\hat{y}_i - y_i)^2\ ]}$$

$$MAPE = (100/n) \sum_i |(y_i - \hat{y}_i) / y_i|$$

- **Cross-Race Panel Regression**

For each driver d, an OLS is fitted on pooled data from 8 prior 2025 races (Australia, China, Japan, Bahrain, Saudi Arabia, Miami, Emilia Romagna, Monaco) with race fixed effects:

$$LapTime = \beta_0 + \beta_1 \cdot TyreLife + \beta_2 \cdot (TyreLife \times \mathbb{1}_\{MED\}) + \beta_3 \cdot (TyreLife \times \mathbb{1}_\{SOFT\}) + \beta_4 \cdot LapNumber + \Sigma_\{r\}\ \delta_r \cdot \mathbb{1}_\{Race=r\} + \varepsilon$$

where δ_r are race fixed effects (one race dropped as baseline). The compound-specific

degradation slopes are:

$$\gamma(d, HARD) = \beta_1$$

$$\gamma(d, MEDIUM) = \beta_1 + \beta_2$$

$$\gamma(d, SOFT) = \beta_1 + \beta_3$$

- **Non-Negativity Correction and Fallbacks**

If γ(d, c) < 0, it is replaced by the compound-level median across drivers:

$$\gamma(d, c) \leftarrow median\{ \gamma(d', c) : \gamma(d', c) > 0,\ d' \in D \}$$

If no driver-specific panel estimate exists for compound c, the same compound median is used.

- **Base Pace Anchored to Montreal**

The base lap time (intercept) is calibrated to Montreal race data for each driver d and compound c:

$$\alpha(d, c) = mean(LapTime_Montreal(d, c)) - \gamma(d, c) \cdot mean(TyreLife_Montreal(d, c))$$

If fewer than 3 Montreal laps exist for a compound, the intercept is estimated using a cross-race offset from a reference compound:

$$\alpha(d, c) = \alpha(d, c_ref) + [\ mean_panel(d, c) - mean_panel(d, c_ref)\ ]$$

where c_ref is an observed compound with a known Montreal intercept and mean_panel denotes the mean lap time from the panel dataset.

- **Expected Lap Time Function**

The predicted lap time passed to the optimizer:

$$T(d, i, c, a) = \alpha(d, c) + \gamma(d, c) \cdot a$$

for all d ∈ D, i ∈ {1,…,70}, c ∈ C, a ∈ {1,…,70}.

## 3. Optimization Model

- **Sets**

*D = {ANT, RUS}* *(drivers)*

*L = {1, 2, …, 70}* *(laps)*

*C = {SOFT, MEDIUM, HARD}* *(compounds)*

*A = {1, 2, …, 70}* *(tire ages)*

- **Parameters**

  $T(d, i, c, a)$ = *predicted lap time*

  $P = 20$ *seconds* (*pit stop penalty*)

  $Inv(d, c, a)$ = *available tire sets*

  $M(SOFT) = 25,\ M(MEDIUM) = 40,\ M(HARD) = 50$ (*max tire life*)

- **Decision Variables**

  $x(d, i, c, a) \in \{0,1\}$ *1 if driver d uses compound c at age a on lap i*

  $y(d, i) \in \{0,1\}$ *1 if driver d pits at end of lap i*

  $u(d, c) \in \{0,1\}$ *1 if compound c is used at least once by driver d*

  $z(d, i, c, a) \in \{0,1\}$ *1 if driver d mounts tire set (c, a) on lap i*

- **Objective Function**

  $$\min Z = \sum_{d} \sum_{i=1}^{70} \sum_{c} \sum_{a} T(d,i,c,a) \cdot x(d,i,c,a) + P \cdot \sum_{d} \sum_{i=1}^{69} y(d,i)$$

- **Constraints**

  ***One tire state per lap*:**

  $$\sum_{c} \sum_{a} x(d, i, c, a) = 1 \qquad \forall\ d, i$$

  ***Lap 1 initialization:***

  $$\sum_{c} \sum_{a} z(d, 1, c, a) = 1 \qquad \forall\ d$$

  $$x(d, 1, c, a) = z(d, 1, c, a) \qquad \forall\ d, c, a$$

  ***Pit stop mount trigger (i = 2,…,70):***

  $$\sum_{c} \sum_{a} z(d, i, c, a) = y(d, i-1) \qquad \forall\ d, i$$

  ***No-pit age progression (a ≥ 2):***

  $$y(d, i-1) = 0 \Rightarrow x(d, i, c, a) = x(d, i-1, c, a-1) \qquad \forall\ d, i \geq 2, c, a \geq 2$$

  ***No fresh tire without pit:***

$$y(d, i-1) = 0 \Rightarrow x(d, i, c, 1) = 0 \qquad \forall\ d, i \geq 2, c$$

***Pit state assignment:***

$$y(d, i-1) = 1 \Rightarrow x(d, i, c, a) = z(d, i, c, a) \qquad \forall\ d, i \geq 2, c, a$$

***Compound usage linking:***

$$\Sigma_{i} \Sigma_{a} x(d, i, c, a) \leq 70 \cdot u(d, c) \qquad \forall\ d, c$$

$$u(d, c) \leq \Sigma_{i} \Sigma_{a} x(d, i, c, a) \qquad \forall\ d, c$$

***Two-compound rule:***

$$\Sigma_{c} u(d, c) \geq 2 \qquad \forall\ d$$

***Inventory constraint:***

$$\Sigma_{i} z(d, i, c, a) \leq Inv(d, c, a) \qquad \forall\ d, c, a$$

***Maximum tire life:***

$$x(d, i, c, a) = 0 \qquad \forall\ d, i, c, a > M(c)$$

***No double-stacking:***

$$y(ANT, i) + y(RUS, i) \leq 1 \qquad \forall\ i = 1, \ldots, 69$$